

NETWORK FUNDAMENTALS

GENNF301

APPLY NETWORK FUNDAMENTALS

Competence

RQF Level: 3

Learning Hours

50

Credits: 5



Sector: ICT and Multimedia

Trade: Computer Systems and Architecture

Module Type: General

Curriculum: ICTCSA3001- TVET Certificate III in Computer Systems and Architecture, Certificate III in Networking and Internet Technologies

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Purpose statement	This module describes the skills, knowledge and attitude required to Apply Network Fundamentals. This module is intended to prepare students pursuing TVET Level 3 in Networking and Internet Technologies. At the end of this module, students will be able to Identify network concepts, apply network protocols and communications and Apply IP addressing (IPv4&IPv6).					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	100%
	Practical work:		70%		70%	
	Group project and presentation	20%				
	Individual project /Work	50%				
			Summative Assessment		0%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Describe network concepts	1.1. Network technologies are properly identified according to their types.
	1.2. Network topologies are properly identified based on their types.
	1.3. Network devices, components are properly identified according to their functions.
	1.4. Data flow is properly identified based on the network models.
2. Apply network protocols and communications	2.1. Network Protocols are correctly classified according to the network layers.
	2.2. Network standards are appropriately applied according to their layers of application.
	2.3 Network media is appropriately applied according to the network types.
3. Apply IP addressing (IPv4 & IPv6)	3.1. IP addresses are correctly classified based on their types.
	3.2. IP addresses are correctly classified based on their versions.
	3.3. IP address structures are appropriately captured based on their versions.
	3.4. IP subnets are appropriately calculated based on the network topology.
	3.5 IP addresses are appropriately assigned according to the network topology.

Course content

Learning outcomes

At the end of the module the learner will be able to:

1. Describe network concepts
2. Apply network protocols and communications
3. Apply IP addressing (IPv4&IPv6)

Learning outcome 1: Describe network concepts

Learning hours: 10

Indicative content

- **Description of network concepts and technologies**

- ✓ Definition of network
- ✓ Network benefits
- ✓ Disadvantages of network
- ✓ Application of network
- ✓ Network classifications
 -  By components roles
 -  By geographical area
- ✓ Network technologies:
 -  IEEE802.3 Ethernet
 -  IEEE802.5 token ring
 -  IEEE802.8 fiber optics
 -  IEEE802.11 Wireless

- **Description of Network topology**

- ✓ Definition of topology
- ✓ Network topology types
 -  Physical
 -  logical
- ✓ Advantages and disadvantages of topology

- **Description of Network components**

- ✓ Media



Physical



logical

- ✓ data
- ✓ Protocols
- ✓ Devices

- **Classification of network devices**

- ✓ Interconnection devices
- ✓ Access devices
- ✓ End devices

- **Description of network models**

Resources required for the learning outcome

Equipment	• Router , Hubs ,Switch , NIC , Computer , Access points, Antenna, Gateways
Materials	Network Cables, internet
Tools	Networking toolkit, simulation tools
Facilitation techniques	▪ Demonstration and simulation ▪ Individual and group work ▪ Research
Formative assessment methods /(CAT)	❖ Written assessment ❖ Oral presentation

Learning outcome 2: Apply network protocols and communications

Learning hours: 20

- **Description of Network Protocols**

- ✓ Definition of network protocol
- ✓ Common network protocols models



NetBEUI



TCP/IP



Apple talk



Novell netware (IPX/SPX)



OSI



CISCO

- **Identification of Network standards**

- ✓ Definition of network standard
- ✓ Importance of standards
- ✓ Internet standards
- ✓ Types of standards



De Facto standards



De Jure standards

- ✓ Standards organizations



ISO



IEEE



ANSI



ITU-Formally CCITT



EIA



telcodia

- **Description of Network Media and Transmission**

- ✓ Major network media types
- ✓ Baseband and broadband transmission technologies
- ✓ Wireless Transmission Techniques

Resources required for the indicative content	
Equipment	Router , Hubs ,Switch ,Computer , Access points, Antenna, Gateways
Materials	Internet bundles
Tools	Simulation tools (eDraw max, cisco packet tracer, GNS 3)
Facilitation techniques	Demonstration and simulation Individual and group work Group discussion
Formative assessment methods /(CAT)	Written assessment Oral presentation

Learning outcome 3: Apply IP addressing (IP v4&IPv6)	Learning hours: 20
Indicative content	
• Description of IP addressing concepts ✓ IP terminologies ✓ IP addressing classification ✓ IP addresses grouping ✓ IP addressing scheme ✓ IP addressing subnet masks ✓ The prefix length • Identification of IP Addresses types ✓ Private ✓ Public ✓ Shared ✓ Dedicated • Application of IPv4 concepts ✓ Introduction to IPv4 ✓ Anatomy of IPv4 address ✓ Methods of Assigning IP addresses  Static  Dynamic	



Automatic

- ✓ Calculation of IP addresses



Binary to decimal conversion



Decimal to binary conversion



Summarization

- ✓ Subnetting



Fixed Length Subnet Masks (FLSM)



Variable Length Subnet Masks (VLSM)



Classless Inter-Domain Routing (CIDR) notation

- ✓ IP addresses Diagnostic tools

- ✓ IP addressing forms



Unicasts



Broadcast



Multicast

- ✓ IP address translation

- **Application of IPv6 concepts**

- ✓ Introduction to IPv6

- ✓ Migration from IPv4 to IPv6

- ✓ Anatomy of IPv6 address

- ✓ Identification of Methods of assigning IP address



Static



Dynamic



Automatic

- ✓ Calculation of IP addresses



Binary to Hexadecimal conversion



Hexadecimal to binary conversion



Shortening of IPv6 Addresses



Summarization

- ✓ Sub-netting



Fixed Length Subnet Masks (FLSM)



Variable Length Subnet Masks (VLSM)



Classless Inter-Domain Routing (CIDR) notation

✓ IP addresses Diagnostic tools

✓ IP addressing forms



Unicasts



Multicast



Anycast

- **Application of IP Configurations**

✓ IPv4 Configuration

✓ IPv6 Configuration

Resources required for the indicative content

Equipment	Router , Hubs ,Switch , Computer , Access points, Gateways
Materials	cables
Tools	calculators, simulation tools
Facilitation techniques	Individual and group work Practical exercise Trainer guided Group discussion
Formative assessment methods /(CAT)	Written assessment Performance assessment

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